

DAI 1000

Digital and AI Literacy

L1 - Course Overview

Teaching Team

Dr. Yongqian LIN 林泳芊

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- Consultation: Email appointment

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Research Interests:

Emerging Technology (VR, AI...); Social Media;
Virtual World; User Experience and Behavior

Facets of the Course

- DAI1000 is one of the University Core courses
 - all students must take and pass
- An introductory course on AI and digital technologies
 - with no pre-requisite required
- There will also be some data analysis practices
- Hopefully, DAI1000 will help you excel in your undergraduate

When to take DAI and how to pass it

Dear Professor Liao,

Hope this email finds you well. My name is [REDACTED] and my student ID is [REDACTED]. I am currently a second-year student in [REDACTED]. I am writing this email to apply for attending your session of [REDACTED].

I am aware that this course is only open to first-year students and could not be registered by sophomores. However, I did not take the course as recommended by the registry because I needed extra credits to attend the [REDACTED] course in the previous academic year.

Since [REDACTED] is a required course for all undergraduates, I hope to have the chance to attend your [REDACTED] course to fulfill the requirements of graduation. I also hope to have the course materials of the first two lectures since students not enrolled in the session do not have the access to course materials on BlackBoard. I

尊敬的廖老师：

您好！

我是来自 [REDACTED]，很抱歉大一上学期最后一节课有事情没来导致了我第一学期 [REDACTED] 成绩不够理想需要重修，因此我特别写这封邮件来请求加课。我保证我这学期绝对 **不会拖欠作业** **保证认真完成小组作业，保证出勤率** 请您批准我能来到 [REDACTED] 的课堂学习，谢谢！

尊敬的教授：

您好！很抱歉打扰您，因大一学年ITE课程结课评价因个人私事 **缺席** 导致教授遗憾未达到修课合格标准，这学期进行课程安排后决定添加重修 [REDACTED] 您任教的时间段，加课系统上我未满足前提条件故而无法通过正式加课进行选课，现提前向您请示可否通过交表添加您的课程，请您有空时回复这封邮件，诚挚感谢

Course Details

- It carries one unit
- One lecture per week
- 14 teaching weeks:
 - 2 assessment sessions (W7-Midterm & W14-Final)

Attendance Policy

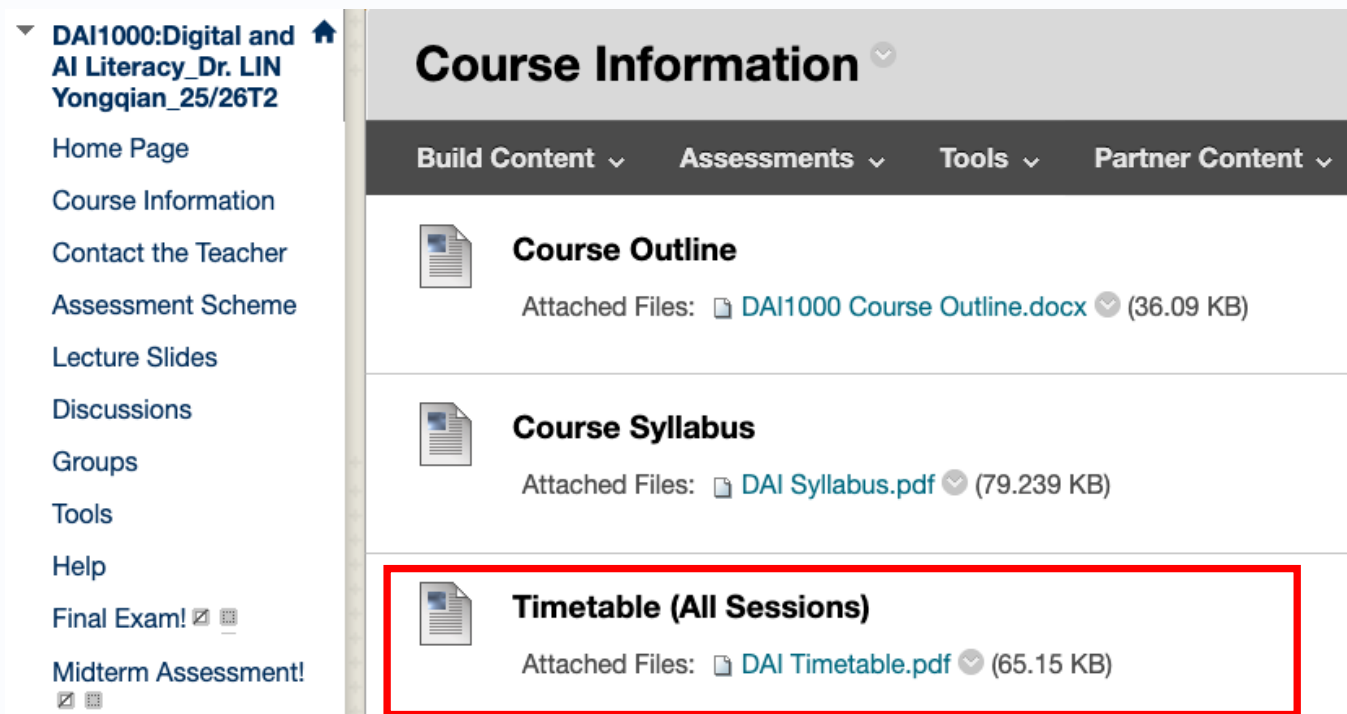
- We will take attendance (**from the third teaching week**)
- You will be given an **F grade** if you fail to attend **three lessons** or more without proper justification
- If you get an F grade, you must **repeat the course until you pass**

Attendance Policy

- We do not accept random request for leave, either verbally or by email
- You need inform me about your absence with valid reasons and supporting documents (e.g., doctor letter) **before the class**
- If not, you will be marked as absent!
- It is your responsibility to inform us about your status

Make-up Classes

- Make-up sessions are possible
- Please email us before attending a make-up class
- Please see the timetable on BB for all sessions



DAI1000: Digital and AI Literacy Dr. LIN Yongqian_25/26T2

Home Page
Course Information
Contact the Teacher
Assessment Scheme
Lecture Slides
Discussions
Groups
Tools
Help
Final Exam! 
Midterm Assessment! 

Course Information

Build Content ▾ **Assessments** ▾ **Tools** ▾ **Partner Content** ▾

 **Course Outline**
Attached Files:  [DAI1000 Course Outline.docx](#) (36.09 KB)

 **Course Syllabus**
Attached Files:  [DAI Syllabus.pdf](#) (79.239 KB)

 **Timetable (All Sessions)**
Attached Files:  [DAI Timetable.pdf](#) (65.15 KB)

These documents provide the course objectives, course syllabus, and grade descriptor

Make-up Classes

TimeTable of DAI1000					
	Monday	Tuesday	Wednesday	Thursday	Friday
8:30			L03 TA 307		
9:00			8:30 - 9:20 LIN Yongqian		
9:30	L06 TA 307		L04 TA 307		
10:00	9:30 - 10:20 LIAO Ruizhi		9:30 - 10:20 LIN Yongqian		
10:30		L08 TA 207			
11:00		10:30 - 11:20 LIN Yongqian			
11:30		L09 TA 207			
12:00		11:30 - 12:20 LIN Yongqian			
13:30		L10 TA 207	L07 TA 307		
14:00		13:30 - 14:20 LIN Yongqian	13:30 - 14:20 LIAO Ruizhi		
14:30	L02 TA 307	L11 TA 207	L01 TA 307		
15:00	14:30 - 15:20 LIAO Ruizhi	14:30 - 15:20 LIN Yongqian	14:30 - 15:20 LIAO Ruizhi		
15:30				L16 TC 208	
16:00				15:30 - 16:20 LIAO Ruizhi	
16:30	L05 TA 107	L15 TA 207			
17:00	16:30 - 17:20 LIAO Ruizhi	16:30 - 17:20 LIN Yongqian			
17:30					

Assessments

- Participation (**10%**): Attendance will be referred
- Tutorials (**20%**): Performance in tutorials
- **Midterm** (**35%**) : Data analysis
- **Final quiz** (**35%**) : Multiple-choice questions

Assessments

- **Midterm (35%)** : Data analysis

Lesson 2 Some Useful Functions

The screenshot displays a video player interface. The main content is a Microsoft Excel spreadsheet. The spreadsheet has the following data:

	A	B	C	D	E
1	Happy day (Y/N)	Day	Apple sales (¥)	Profit (¥)	Rent (¥)
2	yes	1	58	33	5
3	no	2	47	23	
4	deficit!	3	10	-11	
5	deficit!	4	20	-5	
6	yes	5	77	57	
7	yes	6	80	50	
8	yes	7	98	48	
9	yes	8	90	30	
10	no	9	46	16	
11	no	10	63	25	

A blue box with the text "VLOOKUP finds things by a key word" is overlaid on the spreadsheet. The video player interface shows a progress bar at 5:51 / 8:08 and a volume icon.

- Download video file: [2_Useful_Functions.mp4](#)

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Happy day (Y/N)	Day	Apple sales (¥)	Profit (¥)	Rent (¥)								
2	Yes	1	58	33	5								
3	No	2	47	23									
4	Deficit!	3	10	-11									
5	Deficit!	4	20	-5									
6	Yes	5	77	57									
7	Yes	6	80	50									
8	Yes	7	98	48									
9	Yes	8	90	30									
10	No	9	46	16									
11	No	10	63	25									
12													
13	Apple sum:		589										
14	Apple sum(built-in):		589										
15	Minimum:		10										
16	Maximum:		98										
17	Row1, col1:		1										
18	Row3, col3:		-11										
19													
20													
21													
22													
23													
24													

"VLOOKUP"

finds things by
a key word

Assessments

- **Final quiz (35%)** : Multiple-choice questions

QUESTION 12

1 points

A cyber attack in which a swarm of bots controlled by a malicious server attacks a legitimate server to overwhelm its service is called:

- ☐ A spoofing attack
- ☐ A brute force attack
- ☐ A privilege escalation attack
- ☐ A denial of service attack

QUESTION 17

1 points

A student would like to use a picture for her IT video project. How can she do it?

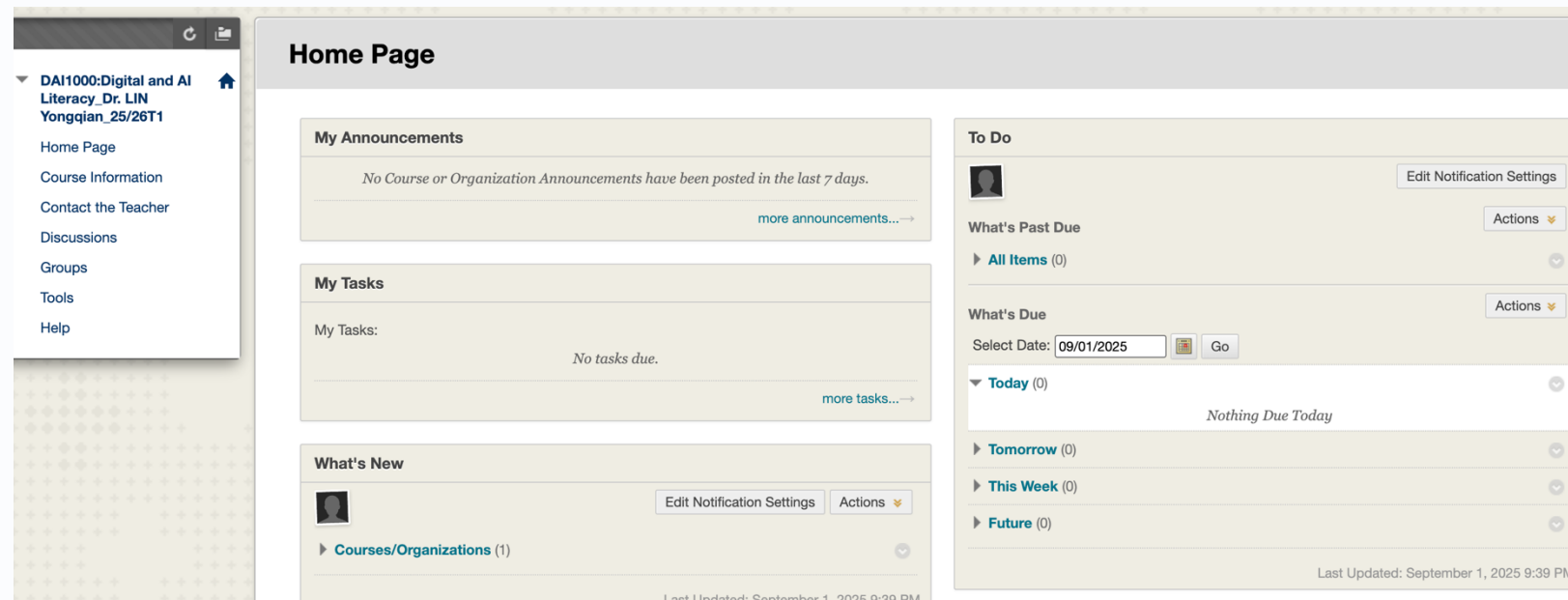
- i. Draw it by herself using Photoshop
- ii. Search on the Internet and copy the picture into her video
- iii. Take a photo of a picture in a magazine and remove the watermark using Photoshop
- iv. Provide a proper citation to the source of the picture at the end of her video

Grade	Description
DISTINCTION	Exceptional performance in all aspects of the course. Demonstrates a deep understanding of how IT and AI technologies transform lifestyles and impact societal development. Shows a strong ability to acquire new knowledge and skills independently. Proficient in using the Internet with excellent etiquette and capable of implementing basic and advanced cybersecurity measures while staying vigilant against threats. Exhibits solid knowledge of information management, ethical principles, and local and international legal and cultural standards. Displays outstanding communication skills and a commitment to leveraging IT and AI responsibly and effectively.
PASS	Good performance in all aspects of the course. Demonstrates a satisfactory understanding of how IT and AI technologies influence lifestyles and societal development. Shows the ability to acquire new knowledge and skills independently. Proficient in using the Internet with acceptable etiquette and capable of implementing basic cybersecurity measures. Possesses adequate knowledge of information management, ethical standards, and local and international legal and cultural norms. Displays good communication skills and a cooperative teamwork spirit.
FAIL	Performance does not meet the minimum requirements of the course's assessment tasks.

If you miss one test, you may easily get FAIL.

Online Resources and Consultation

- Course Materials and e-learning: <https://bb.cuhk.edu.cn>



- Need help with this course? —Contact your lecturer via email

เราแค่พอผ่านได้ (Distinction 4.0-4.9)

Questions?

Three Modules

- History, concept, & related technologies
- Core AI techniques and applications
- AI in Society: ethics, impacts, and future

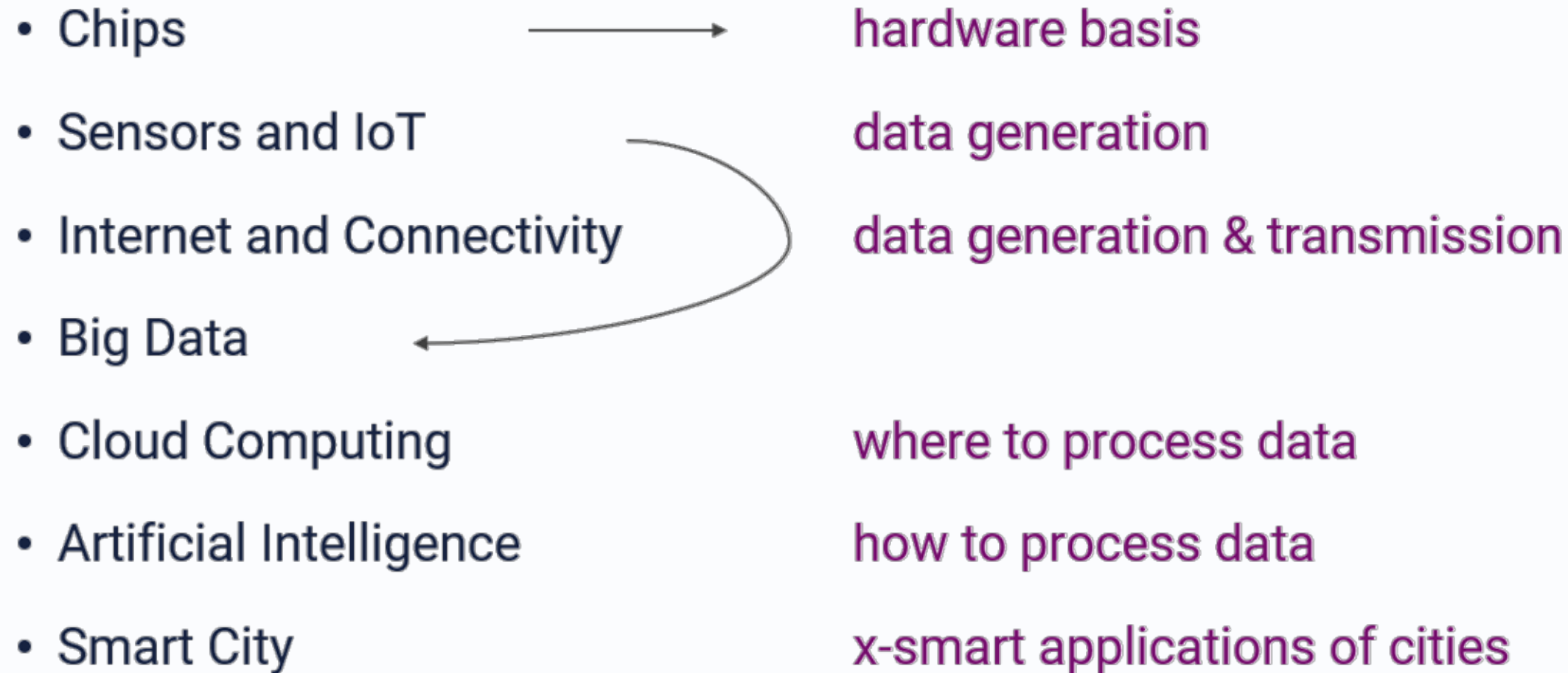
Week	Lecture Content
1	Course Overview
2	Information Technology Developments
3	AI and Technology Convergence
4	Machine Learning Essentials I
5	Machine Learning Essentials II
6	Generative AI and Creative Applications
7	Midterm Assessment
8	Embodied AI and AI Agents
9	AI Agent Practice
10	AI and Cybersecurity
11	AI Ethics and Social Impacts
12	AI Limitation and Trend Discussion
13	Wrap up
14	Final Assessment

Module 1

- Information Technology Developments
 - IoT, Chips, Cloud, big data, and tech evolutions etc
- AI and Technology Convergence I
 - AI history, milestones, and foundational concepts
- AI and Technology Convergence II
 - Evolving technologies that contribute to AI progress

Module 1

- Information Technology Developments



Module 1

- AI and Technology Convergence
 - AI history, milestones, and foundational concepts

AI represents a concerted effort to **simulate human intelligence** in terms of information processes and explore potential ways to **automate these processes** to obtain desirable outcomes.
 - Evolving technologies that contribute to AI progress

AI deals with questions of how to see (vision), move (robotics), communicate (natural language, speech), and learn (memory, reasoning, classification).

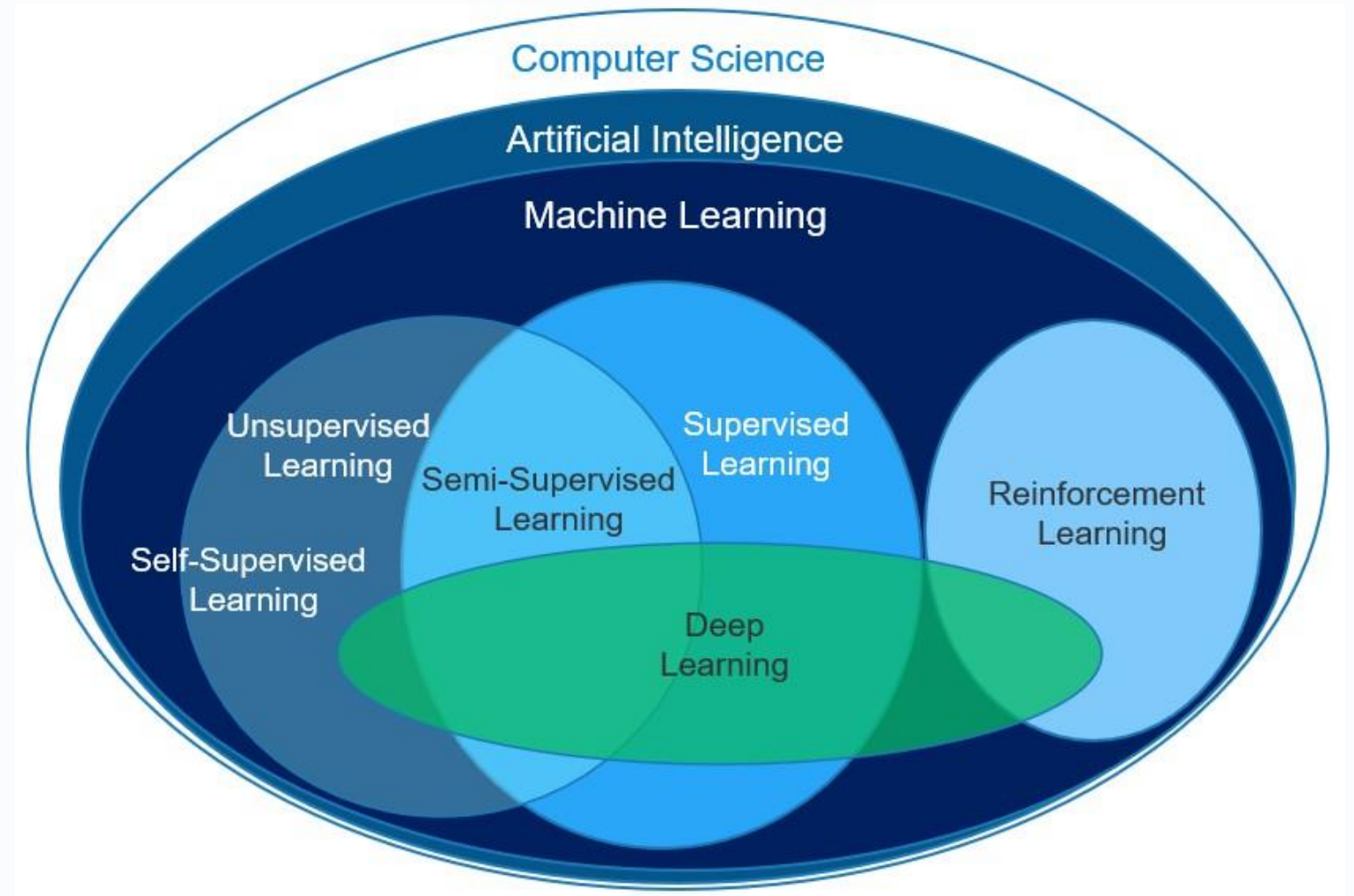
Module 2

- Machine Learning Essentials
 - Different types of machine learning and deep neural networks
- Generative AI and Creative Applications
 - Text, image, video, and copyright issues
- Embodied ^{robot} AI and AI Agents
 - Task automation and AI action in the real world
- AI and Cybersecurity
 - Loopholes, attacks, and defending measures

Module 2

- Machine Learning Essentials

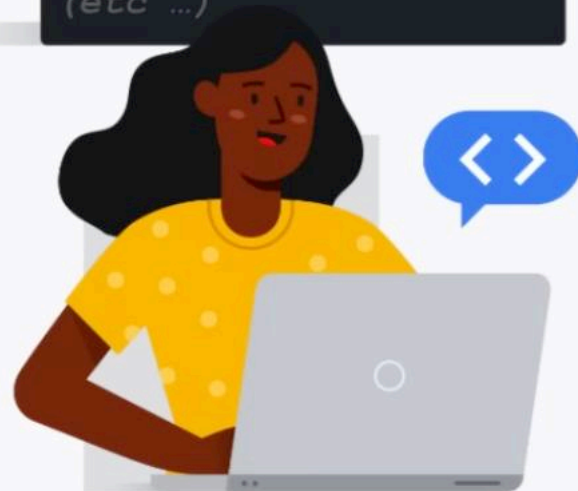
Different types of machine learning
and deep neural networks



Module 2

Traditional programming

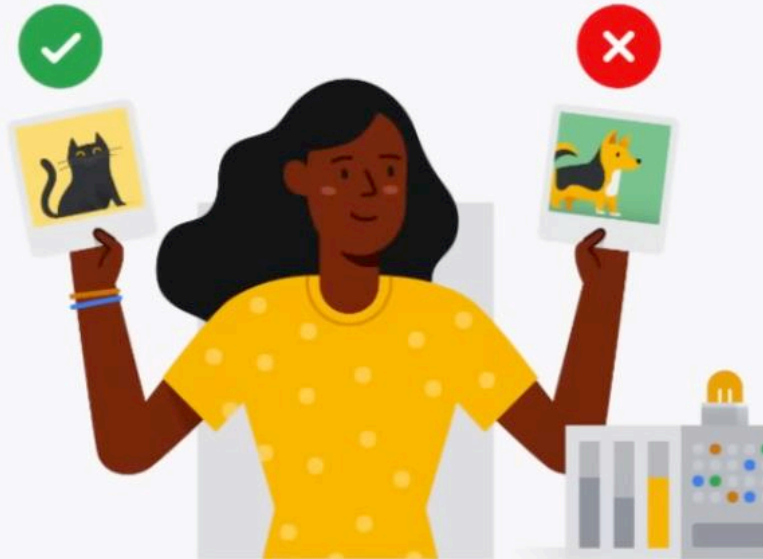
```
Cat:  
  type: animal  
  legs: 4  
  ears: 2  
  fur: yes  
  likes: yarn, catnip  
  (etc ...)
```



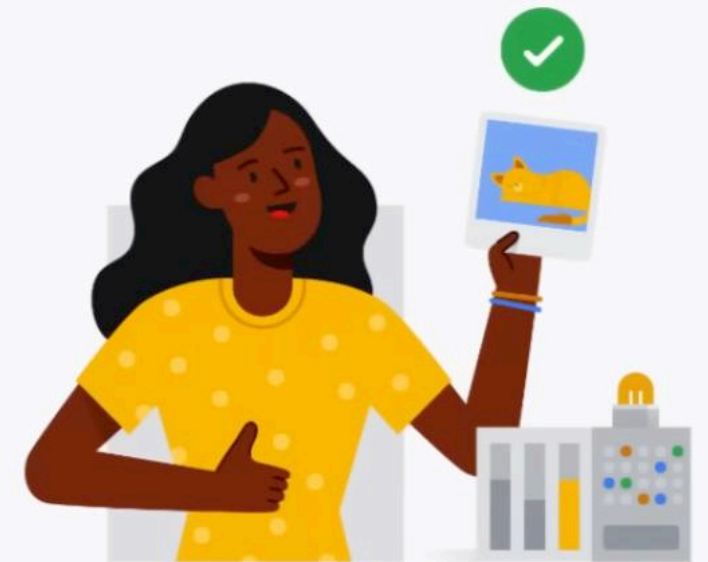
Wave of neural networks | ~2012

This is a cat.

This is not a cat.



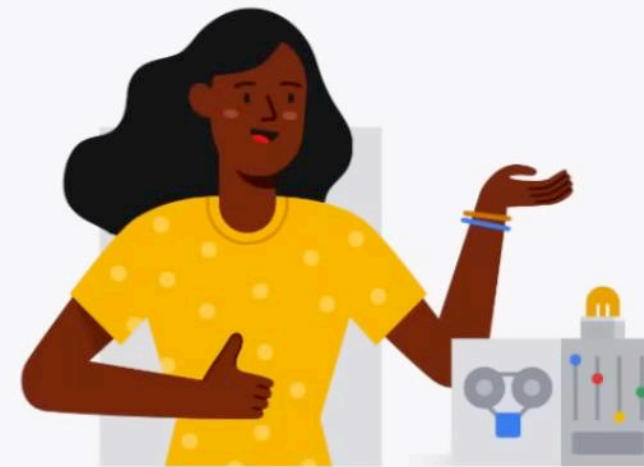
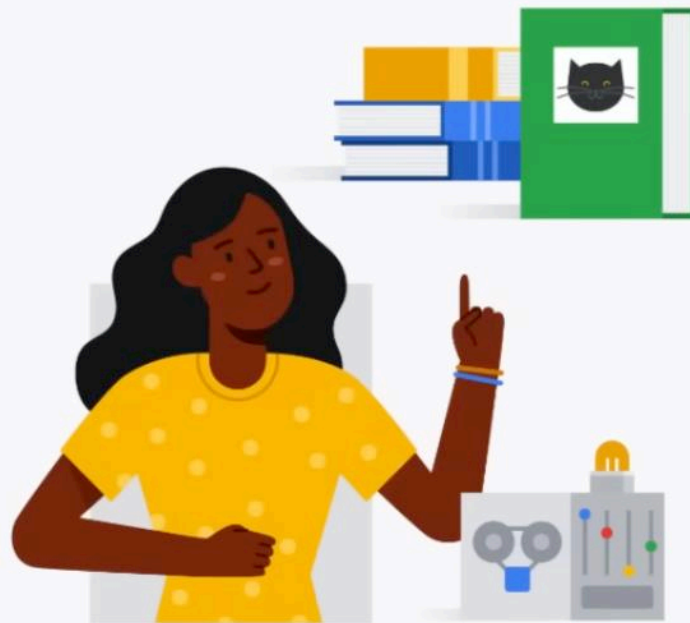
Is this a cat?



Module 2

Generative language models | LaMDA, PaLM, GPT, etc.

Go read this huge pile of books.



So, you've learned about cats and millions of other concepts

What's a cat?

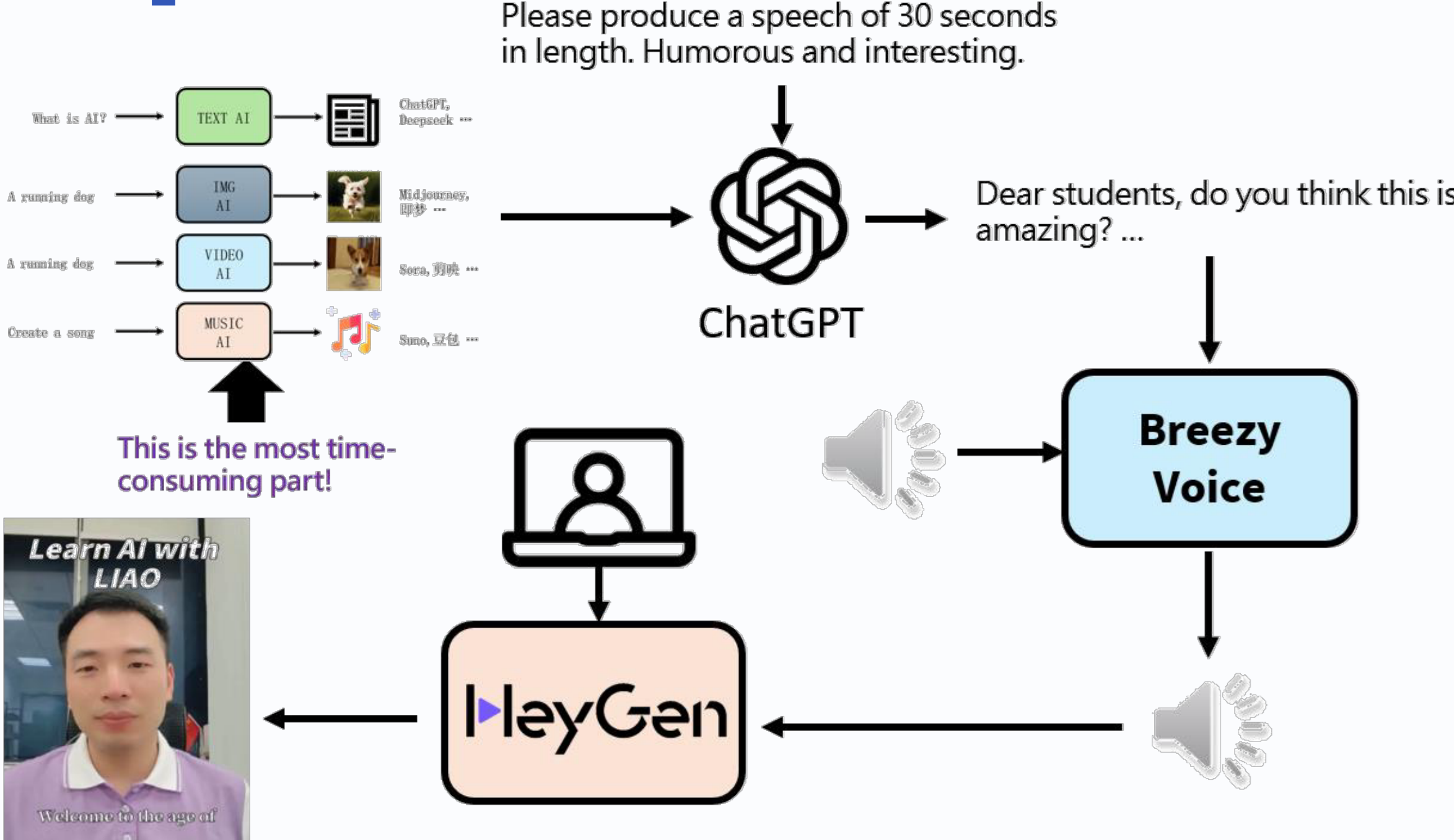
A cat is ...

Module 2

- Generative AI and Creative Applications
 - Text, image, video, and copyright issues



Example



Make teachers obsolete?

帮我生成 PPT：帮我准备一个2小时的讲座PPT。讲座主题是：“语言技术简史与AI新纪元：从规则到生成式革命”。需要回顾语言技术的演进，里程碑，流派，趣闻轶事等；重点讲解语言技术和AI的结合，现状，巨大的潜力，当前的局限性，优缺点，和未来趋势；最后结合2个现场demo，讲解语言技术和AI的结合带来的巨大变化。

已完成思考

接到这个制作 2 小时讲座 PPT 的需求，我先想想可能的情况。提出需求的人大概率是从事教育行业的讲师，或者是科技领域相关企业负责培训、分享的人员。他们的目的很明确，就是要一个能支撑 2 小时讲座的 PPT，主题是“语言技术简史与 AI 新纪元：从规则到生成式革命”，内容要涵盖语言技术演进、与 AI 结合的各方面情况以及现场 demo 讲解等。这个 PPT 的应用场景就是一场专业的讲座，面对的可能是对语言技术和 AI 感兴趣的学生、从业者等。

基于这样的场景，写作风格得专业、严谨，同时又要通俗易懂，毕竟要让听众能理解复杂的技术内容。用词上要偏向技术术语，但也要避免过于晦涩，必要时进行解释说明。

首先，我要回顾语言技术的演进，这部分该从哪里入手呢？语言技术的发展肯定有一个起点，或许可以从早期的简单语言处理开始讲起。那有哪些重要的里程碑呢？比如机器翻译的初步尝试、语音识别技术的突破等。流派方面，规则主义、统计主义、连接主义这些都得涉及，它们各自的特点和发展历程是怎样的呢？趣闻轶事的话，或许可以找一些在技术发展过程中比较有意思的事件，比如某个早期项目的有趣经历。

我先试写一段关于语言技术早期演进的内容：

语言技术的发展最早可追溯到 20 世纪中期，当时人们开始尝试用机器来处理语言。在那个时候，主要是基于规则的方法，研究者们手动编写大量的语法和语义规则，让机器按照这些规则来理解和生成语言。

Help me generate a PowerPoint: Please prepare a PowerPoint presentation for a two-hour lecture on the topic of "A Brief History of Language Technology and the New Era of AI".

The presentation should review the evolution of language technology, including its milestones and schools of thought. The presentation should focus on the integration of language technology and AI, including its current status, enormous potential, limitations, strengths and weaknesses, and future trends.

Module 3

- AI Ethics and Social Impacts
 - Fairness, biases, privacy, and social impacts
- AI Limitations and Trend Discussion
 - Limitations, pitfalls, and future trends of AI

Expectations

- Independent
- Teamwork
 - Critical in the University and workplace
- Punctuality
 - For classes and coursework
- Always check your CUHK email
- Honesty
 - Avoid plagiarism and cheating in quizzes and tests

Honesty in Academic Work

- CUHKSZ places very high importance on honesty in academic work submitted by students, and adopts a policy of **zero tolerance** for cheating and plagiarism
- Any related offence will lead to disciplinary actions, including **termination of studies** at the University

Questions?